

VARANUS ISLAND, Australia

WMO: 953030

Lat: 20.656S

Lon: 115.576E

Elev: 25

StdP: 101.03

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	19.3	6.2	5.9	21.5	7.4	6.4	21.8	17.1	21.2	15.9	20.4	9.1	170	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	4.1	33.4	25.3	32.6	25.6	31.9	25.6	28.8	30.0	28.0	29.8	27.4	29.7	5.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.6	25.1	29.4	27.6	23.6	28.9	26.8	22.5	29.0	94.4	29.9	90.2	29.7	87.4	29.7	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.3	12.8	18.2	36.1	0.3	1.2	18.0	37.0	17.8	37.8	17.6	38.5	17.4	39.4		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.9	28.5	29.0	29.6	28.5	25.7	23.0	21.8	22.4	23.5	25.4	26.5	27.6		
	DBStd	3.00	1.55	1.47	1.27	1.34	1.71	1.42	1.06	1.04	1.27	1.65	1.57	1.51		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	5820	573	533	608	555	487	389	367	386	404	478	496	545		
	CDD18.3	2778	315	299	349	305	228	139	108	127	154	219	246	287		
	CDH23.3	24257	3577	3608	4435	3513	1748	347	76	189	462	1409	2013	2879		
	CDH26.7	7796	1291	1457	1984	1231	250	2	0	2	29	236	460	854		
(14) Wind	WSAvg	7.1	8.0	7.3	6.5	5.9	6.5	7.3	6.8	6.3	7.3	8.0	7.9	7.6		
(15) Precipitation	PrecAvg	272	22	48	52	34	38	41	18	4	1	1	2	7		
	PrecMax	685	100	212	264	282	186	208	131	20	7	26	25	44		
	PrecMin	88	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	157	30	65	70	63	50	47	29	4	2	5	5	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.3	34.7	34.6	32.9	30.6	27.1	25.9	26.7	28.7	31.1	33.3	33.2		
		MCWB	25.0	26.2	25.2	24.7	22.4	20.5	20.1	20.3	20.5	22.7	22.3	24.4		
	2%	DB	32.7	33.1	33.1	32.0	29.6	26.2	24.7	25.7	27.5	29.7	31.0	31.9		
		MCWB	26.4	26.5	25.5	24.1	22.0	20.3	19.2	19.8	20.5	22.8	23.5	24.9		
	5%	DB	31.7	32.2	32.3	31.2	28.8	25.6	24.1	25.1	26.6	28.9	29.9	31.0		
		MCWB	26.3	26.5	25.8	24.1	21.8	19.9	18.8	19.3	20.2	22.5	23.7	24.8		
	10%	DB	30.8	31.4	31.6	30.6	28.0	25.1	23.6	24.4	25.7	28.0	29.0	30.1		
		MCWB	26.2	26.4	25.8	24.0	21.5	19.3	18.4	18.8	19.7	22.0	23.4	24.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.2	29.9	29.7	28.2	25.6	24.4	22.2	23.7	25.2	28.0	28.8	27.4		
		MCDB	30.0	30.8	30.5	30.0	26.9	25.5	23.9	24.5	25.8	28.8	29.4	30.2		
	2%	WB	28.2	28.8	28.5	27.3	24.7	23.1	21.5	22.5	22.8	25.3	26.3	26.7		
		MCDB	30.0	30.1	29.9	29.6	26.8	24.6	23.3	23.9	24.4	26.3	27.9	29.6		
	5%	WB	27.6	27.7	27.7	26.5	24.1	22.3	20.7	21.6	21.8	23.7	25.1	26.1		
		MCDB	29.9	29.9	29.8	29.4	27.0	24.2	23.0	23.3	24.4	27.0	27.8	29.1		
	10%	WB	27.0	27.1	27.1	25.9	23.5	21.5	20.0	20.6	21.0	22.9	24.3	25.6		
		MCDB	29.7	29.9	29.7	29.1	27.0	23.9	22.6	23.0	24.0	26.6	27.5	28.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.4	4.1	3.9	3.4	2.9	3.3	4.1	4.7	5.0	5.0	4.7		
		MCDBR	5.0	5.1	4.8	4.5	4.2	3.4	3.9	4.8	5.5	5.6	5.7	5.5		
		MCWBR	2.2	2.3	2.6	3.0	3.1	2.9	2.8	2.8	3.0	2.8	2.9	2.5		
	5% WB	MCDBR	4.4	4.6	4.5	4.2	3.9	3.0	3.5	4.6	4.9	5.1	5.0	4.6		
		MCWBR	2.2	2.4	2.6	2.9	3.0	2.2	2.7	2.9	2.8	2.9	2.8	2.3		
(40) Clear-Sky Solar Irradiance	taub		0.400	0.397	0.374	0.353	0.335	0.325	0.316	0.315	0.333	0.353	0.357	0.373		
	taud		2.539	2.563	2.623	2.644	2.638	2.652	2.657	2.619	2.564	2.529	2.549	2.572		
	Ebn at Noon		945	937	934	916	896	887	908	941	959	967	980	972		
	Edh at Noon		111	107	97	88	82	78	80	89	101	109	109	108		
(44) All-Sky Solar Radiation	RadAvg		7.66	7.35	6.54	5.59	4.65	4.18	4.56	5.67	6.73	7.61	8.14	8.18		
	RadStd		0.43	0.49	0.44	0.36	0.38	0.31	0.32	0.17	0.26	0.14	0.17	0.28		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page